DATA ANALYTICS

Introduction to Analytics

Turning Data into Insight. Insight into Action. Action into Value.

Analytics is the process of collecting, transforming, and organizing data to identify patterns, draw conclusions, and make informed decisions. It involves a range of tools and techniques to turn raw data into actionable insights that can improve business performance, optimize processes, and predict future trends.

> Data is the new oil, but analytics is the engine that turns it into power.
>
> — Unknown

Overview | Types (D2P3) | Patterns | Tools & Data | Analysis vs Science | Practice & Quiz

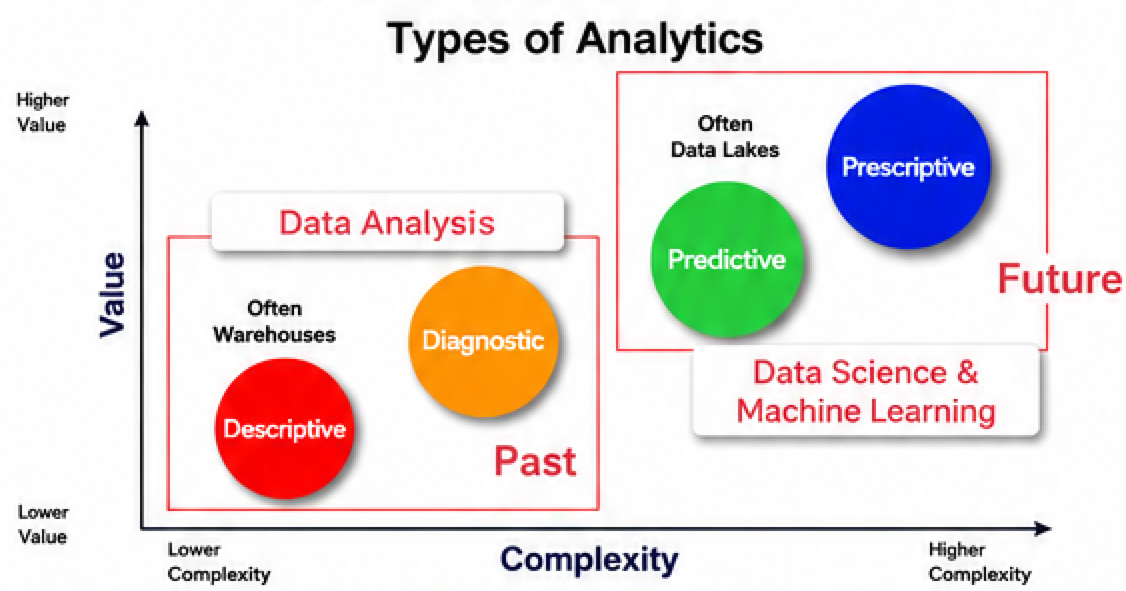
1. ANALYTICS AT A GLANCE

→ **2. Collect** Gather data from relevant sources. → **3. Clean & Transform** Remove errors, handle missing data, and organize for analysis. → **4. Analyze** Apply techniques and models to find patterns and insights. → **5. Share & Act** Communicate insights and drive action for better outcomes.

Data cleaning is a critical step to remove duplicates, errors, and missing information before analysis can begin. Techniques include statistical analysis, regression, time series, data mining, and machine learning.

2. D2P3 FRAMEWORK: THE FOUR TYPES OF ANALYTICS

Analytics evolves from understanding the past to shaping the future.



Descriptive Analytics
What happened?
Summarizes past data to understand events and performance.

Diagnostic Analytics
Why did it happen?
Investigates root causes behind past events.

Predictive Analytics
What will happen?
Uses historical data and models to forecast future outcomes.

Prescriptive Analytics
What should we do?
Recommends the best actions to achieve desired outcomes.

Purpose: Move from reporting what happened, to understanding why, to predicting what's next, to prescribing what to do.

3. COMMON PATTERNS IDENTIFIED IN ANALYTICS

1. Trend Analysis
- Upward trend
- Downward trend
- Horizontal trend

Example: Sales increasing quarter over quarter.

2. Seasonal & Cyclical
- Seasonality: repeats over fixed intervals
- Cyclicality: longer, less predictable cycles

Example: Retail spikes in holiday season.

3. Correlation & Relationships
- Positive correlation
- Negative correlation
- No correlation

Example: Temp ↑ → Ice cream sales ↑

4. Clustering & Segmentation
- Group similar data points
- Enables market segmentation

Example: Group customers by purchase behavior.

5. Anomalies & Outliers
- Detect unusual deviations
- Useful for fraud & monitoring

Example: Unusual credit card transaction amount.

6. Association & Frequent Itemsets
- Finds items occurring together
- Market basket & web usage mining

Example: Milk and bread bought together.

See examples and try it yourself in the interactive demo below

4. INTERACTIVE DEMO: EXPLORE PATTERNS IN YOUR DATA

Choose Pattern: Trend Analysis
Select Metric: Sales
View: Table | Chart

Try different patterns and metrics to explore insights.

Month	Sales	Units	Returns
Jan	5,230	165	108
Feb	6,120	198	103
Mar	6,980	220	112
Apr	6,450	210	115
May	7,550	245	109
Jun	8,430	268	123
Jul	8,900	290	127
Aug	9,220	305	131
Sep	9,800	322	129
Oct	10,300	341	155
Nov	11,200	362	140
Dec	10,400	333	136

Quick Insights

Overall upward trend
Sales increased 98.9% from Jan to Nov.

Seasonal effect
Slight dip in Dec after Nov peak — common in retail.

Watch returns
Returns rise slightly in the second half of the year.

- **Data Science** Broadest field using data to solve complex problems.
- **Data Analysis** Focuses on extracting meaning from data in front of you.
- **Data Analytics** Turns data into insights to improve decisions and performance.
- **Data Mining** Finds hidden patterns and relationships in large datasets.

5. TOOLS I KNOW

- Microsoft Excel & Access
- Python & R
- Power BI & Tableau
- Minitab (Statistical Analysis)

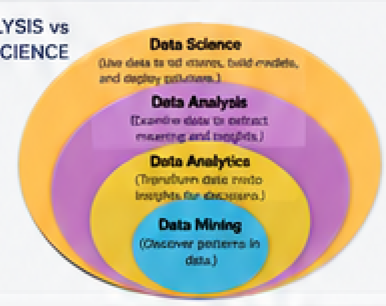
6. DATA ORGANIZATION & TRANSFORMATION

Structured Data
(SQL Database, Wonderware, PostgreSQL)
Organized in rows and columns. Easy to store, query, and analyze.

Unstructured Data
(MongoDB)
No predefined format. Requires special tools to process.

Big Data
(APACHE SPARK, APACHE HADOOP, etc.)
Spark
Large volume, velocity, variety. Requires distributed processing.

7. ANALYSIS vs DATA SCIENCE



KEY TAKEAWAY Analytics helps us move from data to decisions. The right patterns, tools, and understanding create real business impact.

REMEMBER Good analytics answers the right questions, with the right data, using the right methods, and drives the right actions.

NEXT STEPS Explore the demo, test your knowledge in the quiz, and download the posters to reinforce learning.

Descriptive: What happened? **Diagnostic:** Why did it happen? **Predictive:** What will happen? **Prescriptive:** What should we do?